Pricing of a Multi-Asset Bermudan Worst-of Put with Heston Stochastic Volatility

ENSF 620 Project Proposal

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Introduction

An essential capability in quantitative finance is conducting the analysis of the fair price of an asset or derivative. On the buy side, understanding fair pricing backed by statistical analysis allows a trading desk to view what is undervalued or overvalued in the market, who can therefore buy and sell those financial products accordingly with statistical edge. In the publicly traded market, the value of an asset or derivative is highly liquid, allowing its actual price to be determined by supply and demand. However, certain over-the-counter (OTC) derivatives exist that are not publicly traded, and are therefore illiquid, meaning there is no observable market data to determine the price of that contract. On the sell side, financial institutions selling these customized contracts must therefore rely on internal statistical models to determine the fair price of that contract so they can hedge their risk. This project will develop a framework to run this type of stochastic computational analysis to determine the fair price of a complex and customized contract that is illiquid and not publicly tradable. The case study used will be a worst-of multi-asset Bermudan put option.

The payoff of a worst-of equity derivative depends on the worst-performing asset in a correlated basket. The worst-of puts embedded in retail structured products are frequently European, meaning they can only be exercised at their expiry date, T. This project instead studies a Bermudan (discretely exercisable at $t_k < T$) worst-of put as a representative multi-asset payoff with early exercise, which introduces an optimal-stopping problem that must be solved numerically through simulation of the underlying assets. Pricing it requires the Least-Squares Monte Carlo method (LSMC) [10, 7], the standard simulation method for early-exercise contracts. The assets will be simulated following the Heston stochastic volatility model [8], which is widely used because it produces a volatility skew and links falling prices to rising volatility through the correlation ρ, both of which are observed in real market data.

A model is only as good as its fit to the market it is meant to price into. Rather than assume parameter values, each asset's Heston parameters are calibrated by least squares to that asset's own listed European volatility surface, using a grid of strikes and maturities chosen to include the worst-of contract's own strike and expiry. This project builds and verifies the calibration, simulation, and optimal exercise policy, and reports the price of the worst-of Bermudan put together with the value of its early-exercise feature relative to the corresponding European contract.

A. Question, motivation, and scope

Primary question. For a three-asset Bermudan worst-of put, what is the risk-neutral price implied by Heston parameters calibrated to each asset's own listed European volatility surface? How much of that price reflects the value of early exercise, relative to the corresponding European worst-of put?

Motivation. A financial institution cannot quote an OTC exotic contract without a modelled price backed by statistical analysis to hedge their own risk. That price depends on two things: parameters that reflect the market's own view of volatility, and a numerical method that correctly handles the worst-of payoff's early-exercise. This project builds and verifies that pipeline and reports how much the early-exercise feature is worth.

Scope. The work is tiered for a minimum defensible result:

- **Tier 1 (minimum defensible result):** a single-asset Heston Bermudan put, with parameters calibrated to that asset's own listed volatility surface.
- **Tier 2 (primary target):** the three-asset worst-of put, with all three assets calibrated to their own volatility surfaces.
- **Tier 3 (stretch):** an OpenMP parallelization of the C++ pricing engine

Out of scope: Live data (only a single snapshot is used for calibration Heston parameters); P&L simulation; stochastic rates, jumps, discrete dividends; continuous (true American) exercise.

B. Data/workload and authority

This project uses three kinds of external input. (1) Listed European-option quotes for three chosen underlyings, downloaded once as a fixed snapshot on a stated calibration date, used to fit each asset's Heston parameters to its own volatility surface. (2) Daily closing prices for the same three underlyings over a trailing window, used to estimate cross-asset correlations. (3) Parameter values from published papers, used as a fallback if the market-data calibration is not viable, and as reference values for the verification checks in Table 1. Software dependencies are open-source under permissive licenses (Eigen, pybind11, NumPy, SciPy). No personal or restricted data are involved. The option quotes and price history are public and pertain to traded instruments, not individuals.

The raw snapshot of market data is committed as a .csv file alongside the code for reproducibility. If the snapshot is unavailable, too sparse to identify parameters, or its license is unclear, this project falls back to published Heston parameter values (Table A1). The complete governance worksheet is Table A1 (Appendix).

C. Method, baseline, and computing plan

Model and contract. Under the risk-neutral measure, for assets $i = 1, \dots, d$:

$$dS_{i,t} = (r - q_i)\, S_{i,t}\, dt + \sqrt{\nu_{i,t}}\, S_{i,t}\, dW^S_{i,t}, \tag{1}$$

$$d\nu_{i,t} = \kappa_i(\theta_i - \nu_{i,t})\, dt + \xi_i \sqrt{\nu_{i,t}}\, dW^\nu_{i,t}, \tag{2}$$

$$d\langle W^S_i, W^\nu_i \rangle_t = \rho_i\, dt, \qquad d\langle W^S_i, W^S_j \rangle_t = R_{ij}\, dt. \tag{3}$$

Each asset has five parameters $\phi_i = (\nu_{i,0}, \kappa_i, \theta_i, \xi_i, \rho_i)$, which are fit to market data for each asset. The risk-free rate $r = 5\%$ and dividend yields $q_i = 0$ are treated as known. The contract pays $\left(K - \min_i S_{i,t_k}/S_{i,0}\right)^+$ at the exercise date $t_k \in \{t_1, \dots, t_M\}$ chosen by the holder.

LSMC. At each exercise date, discounted realized cash flows of in-the-money paths are regressed on a polynomial basis in the asset paths and their variances. Variance must be part of the state because the continuation value depends on it in the Heston model. [7].

Initial parameterization. Each asset's Heston parameters ϕ_i are calibrated by least squares to that asset's own market-implied-volatility surface,

$$\hat{\phi}_i = \arg\min_{\phi_i} \sum_{(K,T)\in\mathcal{G}_i} \Big(\sigma_{\text{iv}}^{\text{mkt}}(K,T) - \sigma_{\text{iv}}^{\text{Heston}}(K,T;\phi_i)\Big)^2, \tag{4}$$

where $\mathcal{G}_i$ is a small grid of listed strikes and maturities for asset i that includes the worst-of contract's own (K,T), mid-quotes are converted to Black–Scholes implied volatility, and $\sigma_{\text{iv}}^{\text{Heston}}$ comes from inverting the Heston European price, evaluated by the numerically stable form of the characteristic function [8, 1]. Eq. (4) is minimized with SciPy `least_squares`) from several starting points, with $\nu_{i,0}, \kappa_i, \theta_i, \xi_i > 0$ and $|\rho_i| < 1$. The calibration date and quote snapshot are fixed and recorded in the run manifest, so the fit is reproducible even though the market itself is not static. Cross-asset correlations R_{ij} are the sample correlations of the three assets' trailing daily log-returns over the same window. If a usable quote grid cannot be obtained for an asset, or the fit is unstable across starting points, that asset's parameters fall back to published values from cited papers, and the substitution is stated in the report.

Computing plan. Tier 1 is prototyped in Python (NumPy/SciPy) for correctness. The pricing engine is then ported to C++ and exposed to Python through pybind11. Python handles the calibration, correlation estimation, and figures. With the OpenMP parallelization (Tier 3), this code can be ran with HPC resources which are provided by the Digital Research Alliance of Canada (Trillium, Rorqual, Nibi supercomputers).

D. Repository and reproducibility plan

```
heston-uq/
  README.md  AI_USE.md  CMakeLists.txt
  cpp/{include,src,tests}/       pricing engine
  python/                        calibration, correlation estimation,
                                 analysis
  data/                          market-data snapshot (option
                                 quotes, price history)
  results/                       simulation results and initial conditions
  prototype/                     exploratory Jupyter notebooks (not part of the
                                 assessed workflow)
```

- **Environment.** C++ with CMake compiler and library versions are recorded in README.md
- **Provenance.** Every run writes a log output file with the commit hash, master seed, environment, and the market-data snapshot's date, so a fitted price is traceable to the exact quotes it came from.
- **Randomness.** A seeded random number generator makes results identical regardless of thread count or scheduling. Python and C++ are compared on a shared fixture of pre-generated normals, which provides a second, independent implementation.
- **Reviewer path.** reproduces a reduced Tier 1 result without HPC resources, the README documents the full run instructions.
- **Final version.** An annotated tag `v1.0-final`, submitted by Route A as a ZIP preserving `.git`, verified by extracting it to a temporary directory and running `git status`, and `git log`.

Table 1: Verification checks and pass rules, fixed before implementation. SE = Monte Carlo standard error.

ID	Check	Pass rule
V1	Heston European Monte Carlo vs. analytical solution, 5 maturities × 8 strikes	$\geq 95\%$ of options within 3 SE
V2	Calibration recovers a synthetic surface generated from known Heston parameters	Implied-vol residual $\leq 10^{-4}$
V3	Calibration on the real quote grid, per asset	Implied-vol residual ≤ 0.01
V4	Single-asset Heston Bermudan put, with exercise on T it reproduces the closed-form European put	Difference within 3 SE
V5	Three identical assets on identical Brownian increments vs. single-asset Bermudan price	Difference within 3 SE
V6	Run pricing analysis for 1) Worst-of Bermudan, 2) Worst-of European, 3) 3 single-asset Bermudans	Worst-of Bermudan $\geq$ worst-of European $\geq$ each single-asset Bermudan put
V7	C++ engine vs. Python prototype on the shared fixture	Relative difference $\leq 10^{-10}$
V8	Reproducibility: price from 2 independent master seeds	Seeds agree within 1 SE
V9	Output completeness	All runs present and finished

E. AI/agent plan

AI tools will be used as assistants for understanding the mathematics, identifying bug fixes, and building the framework for the code. Material use and any rejected or corrected suggestions are logged in `AI_USE.md` as they occur and disclosed in the final report. The complete AI-use worksheet is Table A3 (Appendix).

F. Milestones, risks, fallback, and expected evidence

Table 2 lists dated milestones with their pass rules and fallbacks, and Table 3 lists the principal risks with their triggers. The tiers in Section A define the fallback scope: each later tier is attempted only once the previous one passes its checks.

Expected evidence. (E1) Benchmark table for V1–V10 with standard errors; (E2) convergence plots in paths, sub-steps, exercise dates, and basis degree, and an engine comparison (Python vs. C++); (E3) the fitted volatility surface against market quotes for each asset, with fit residuals (V12) and the parameter-recovery check (V11); (E4) the final three-asset worst-of Bermudan price with its early-exercise premium relative to the European worst-of put (V7); (E5) a seed-robustness table (V13); (E6) test and CI logs with run manifests; (E7, stretch) the OpenMP scaling plot and the calibration-grid robustness check. A finding that the early-exercise premium is small, or that a restricted calibration grid gives materially different parameters, is a valid result and will be reported as such.

References

[1] Albrecher, H., Mayer, P., Schoutens, W., & Tistaert, J. (2007). The little Heston trap. *Wilmott Magazine*, January, 83–92.

[2] Andersen, L., & Broadie, M. (2004). Primal-dual simulation algorithm for pricing multidimensional American options. *Management Science*, 50(9), 1222–1234.

[3] Clarke, N., & Parrott, K. (1999). Multigrid for American option pricing with stochastic volatility. *Applied Mathematical Finance*, 6(3), 177–195.

Table 2: Dated milestones. Check IDs refer to Table 1.

Date	Evidence	Pass rule	Trigger → fallback
Sep 29	Proposal	All rubric fields addressed	–
Oct 9	Python prototype: GBM Bermudan put, Heston European MC and closed form; runtime benchmarks	V1, V2, V3	Failing on Oct 16 → no C++ work until fixed
Oct 16	Market-data snapshot acquired for the three chosen assets (option chain and price history); calibration code implemented and checked on synthetic ground truth	V11	Snapshot unusable or licence unclear → published-parameter fallback (Table A1)
Oct 23	Tier 1: single-asset Heston Bermudan put priced with market-calibrated parameters	V4, V5, V12	Fit unstable or RMS residual above threshold → fix κ at a literature value and report it as an assumption
Oct 30	Executable baseline: C++ kernel with pybind11, CI, Tier 1 pipeline	V9, V10; CI passing	C++ failing V9 → Numba engine for production; C++ becomes stretch
Nov 3	Checkpoint: Tier 1 complete and reviewable repository	Smoke test passes from a clean clone	Revise method or evidence plan
Nov 14	Correlation estimated from historical returns; all three assets calibrated; two-, then three-asset worst-of pricing	V6, V7, V8	Failing → two-asset basket as Tier 2
Nov 20	Stable Tier 2 result: worst-of price, early-exercise premium, seed robustness, output completeness	V7, V13, V14	Unstable → freeze Tier 1 (or two-asset) as the main result
Nov 27	Draft report and slides	Every figure regenerated from manifests; every claim mapped to a run	Remove unsupported claims
Dec 1	Presentation	Finishes within the assigned slot	–
Dec 3	Final package	Route A ZIP verified in a temporary directory; smoke test passes	Submit verified minimal package before extras
Dec 4	D2L submission	Uploaded files opened and checked	–

Table 3: Principal risks, triggers, and fallbacks.

Risk	Trigger	Fallback
Scientific: LSMC basis inadequate	V5 fails (price moves > 2 SE between basis degrees)	Add features (e.g., second-worst performance) or reduce the state; report residual bias
Data: market-data snapshot unusable or licence unclear	Cannot obtain a clean grid across the needed strikes/maturities by Oct 16, or redistribution terms are unresolved	Published Heston parameter values from cited papers; state which assets were affected
Calibration: parameters not identifiable or fit unstable	V11 fails on synthetic recovery, or V12's RMS residual is above threshold / unstable across starting points	Fix a poorly-identified parameter (e.g., κ_i) at a literature value and calibrate the rest; state it as an assumption
Computing: pricing too slow	One three-asset pricing takes > 30 s in the production engine, or the full convergence sweep takes > 6 h	Reduce path counts or sweep size, or reduce to two assets
Software: C++ port slips	V9 or V10 failing on Oct 30	Numba engine becomes production; C++ becomes stretch
Schedule: Tier 2 unstable	V6, V7, or V8 failing by Nov 20	Tier 1 becomes the main result

[4] Cox, J. C., Ross, S. A., & Rubinstein, M. (1979). Option pricing: A simplified approach. *Journal of Financial Economics*, 7(3), 229–263.

[5] Fang, F., & Oosterlee, C. W. (2008). A novel pricing method for European options based on Fourier-cosine series expansions. *SIAM Journal on Scientific Computing*, 31(2), 826–848.

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[9] Joshi, M. S. (2008). *C++ Design Patterns and Derivatives Pricing* (2nd ed.). Cambridge University Press.

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Appendix: Planning worksheets

Table A1: Data-governance worksheet.

Field	This project
Source, owner/steward, version/access date	(1) Listed European-option mid-quotes for three chosen underlyings, from [TODO: name the data provider (e.g., a free public source such as a brokerage's option-chain download, or a licensed terminal accessible through the university)], a single fixed snapshot dated [TODO: fix a calibration date]. (2) Daily closing prices for the same three underlyings over the trailing year, from [TODO: same or another public source], for correlation estimation. (3) Published Heston parameter values and benchmark prices from cited papers (calibration fallback; V4/V8 benchmarks). (4) Open-source dependencies, versions pinned in the lock file.
How access is obtained	The option and price snapshots are downloaded once (not a live feed) via [TODO: the chosen method, e.g., a public data download]; papers through the University of Calgary library; dependencies from public package indexes.
Authority for this project use	Own work; public market quotes used for a personal, non-commercial academic exercise – [TODO: confirm the chosen provider's terms of use permit this before submission]; published values used as cited factual inputs; dependencies used under their licences. Technical access to a data source does not by itself establish authority to use it.
Limits: ethics, consent, contracts, licences	No human subjects or personal data. Market-data limits: [TODO: confirm the provider's redistribution terms], especially before any public release. Dependency licences (MPL-2.0, BSD-3-Clause) permit use and redistribution with notices.
Classification and sensitive elements	Public; the option quotes and price history concern publicly traded instruments, not individuals; no identifiable or sensitive elements.
Approved storage and processing locations	Personal laptop; GitHub repository; D2L for submission.
Approved external services	GitHub (hosting, CI); Claude via claude.ai (code and mathematics only; no credentials shared); [TODO: the chosen market-data provider] for the one-time snapshot download.
Repository inclusion/exclusion rule	Include code, configurations, the committed market-data snapshot (option quotes and price history – unlike simulated paths, this cannot be regenerated from a seed), manifests, small summary CSVs, and figures. Exclude simulated paths, per-draw arrays, build directories, virtual environments, and caches.
Retention and deletion	The market-data snapshot is retained indefinitely alongside the code, since it is required to reproduce the calibrated result. Large simulated outputs are deleted once summaries and manifests are verified; they are regenerable from configuration and seed.
Sharing/publication limit	Repository private during the term. Before any public release after final grades, [TODO: re-confirm] that the committed snapshot may be redistributed under the provider's terms; if not, the snapshot is scrubbed from the public release and the published-parameter fallback stands as the publicly reproducible path.
Synthetic/public fallback	If the market-data snapshot cannot be obtained, is too sparse to identify parameters, or its licence is unclear, calibration falls back to published Heston parameter values from cited papers; the substitution is stated in the report.
Pending approvals and stop condition	No dual-instructor approval is required (this project is scoped to ENSF 620 only; a related, separate project in ENME 618 uses this pricer but shares no submitted work product with it). Open item: confirm the market-data provider's terms of use before the Nov 3 checkpoint; if unresolved by then, proceed on the published-parameter fallback and note the substitution.

Table A2: Computing and fallback worksheet. Runtimes are estimates to be benchmarked by October 9.

Question	Plan
Workload that must run	One least-squares Heston calibration per asset (a fast semi-analytic pricer, a few hundred evaluations per fit, well under a second each); a convergence/comparison sweep of LSMC repricings over path count, sub-steps, exercise-date count, basis degree, and engine (a few dozen configurations). No nested outer Monte Carlo loop over parameter draws is required.
Smallest meaningful case	Tier 1 single asset at baseline path count (2×10^4 paths per set); the reviewer smoke test, about 10 minutes on a laptop.
Data size, memory, runtime, repetitions	One three-asset pricing with 10^5 paths per set (two sets), 50 exercise dates, and 4 sub-steps is about 2.4×10^8 state updates, estimated at 1–5 s single-threaded in C++ (a few minutes in the Python prototype). The full convergence/comparison sweep (a few dozen configurations, median over 5 repeats each) is estimated at under an hour on a laptop. The Tier 3 OpenMP scaling study repeats a handful of configurations across 1, 2, 4, 8 threads.
Local baseline environment	Personal laptop [TODO: CPU model, core count, RAM, OS].
Proposed cluster, cloud, or accelerator	None requested; the workload fits a laptop.
Availability, account, quota, or cost risk	None; no external compute accounts are needed. The one external dependency is the market-data snapshot, downloaded once (Table A1).
Rule for comparing environments	Fixed configuration and seed; median wall time over 5 repeats; thread count reported.
Correctness across environments	V9 (C++ vs. Python), V10 (thread invariance), V11 (calibration parameter recovery), smoke result reproduced from a clean clone.
Output completeness	`verify_outputs.py` checks that all runs are present and finite and that manifest hashes match (V14).
Local fallback and trigger	See Table 3 (computing and data rows).

Table A3: AI-use and provenance worksheet (plan; updated as a final disclosure in `AI_USE.md`).

Field	Plan
Tool/model and access mode	Claude (Anthropic) through the claude.ai chat interface and Claude Code through VSCode.
Main roles	Explaining mathematics and algorithms (SDE discretization, LSMC, Heston calibration); helping identify bugs during development; reviewing drafts of the proposal and report.
Actions the tool may take	Produce suggestions in chat, which I read, modify, and test myself.
Out-of-bounds actions and data	Repository write access; generating the reference values used in Table 1; writing reported results, figures, or claims; access to credentials.
Human-written requirements and acceptance criteria	Function specifications and all pass rules in Table 1, written before implementation.
Verification not relying on generated claims	Closed forms, published benchmarks, an independent second implementation, the parameter-recovery check (V11), and convergence/invariance checks; derivations checked against cited sources.
Important rejected or corrected suggestions	Logged in `AI_USE.md` as they occur.
Data/secret boundary	No restricted data exist; no credentials or tokens are shared with AI services or committed.
Residual limitations	Tests with loose tolerances could miss subtle numerical errors, mitigated by convergence studies and multiple independent checks. The market-data snapshot's licence terms are still to be confirmed (Table A1).